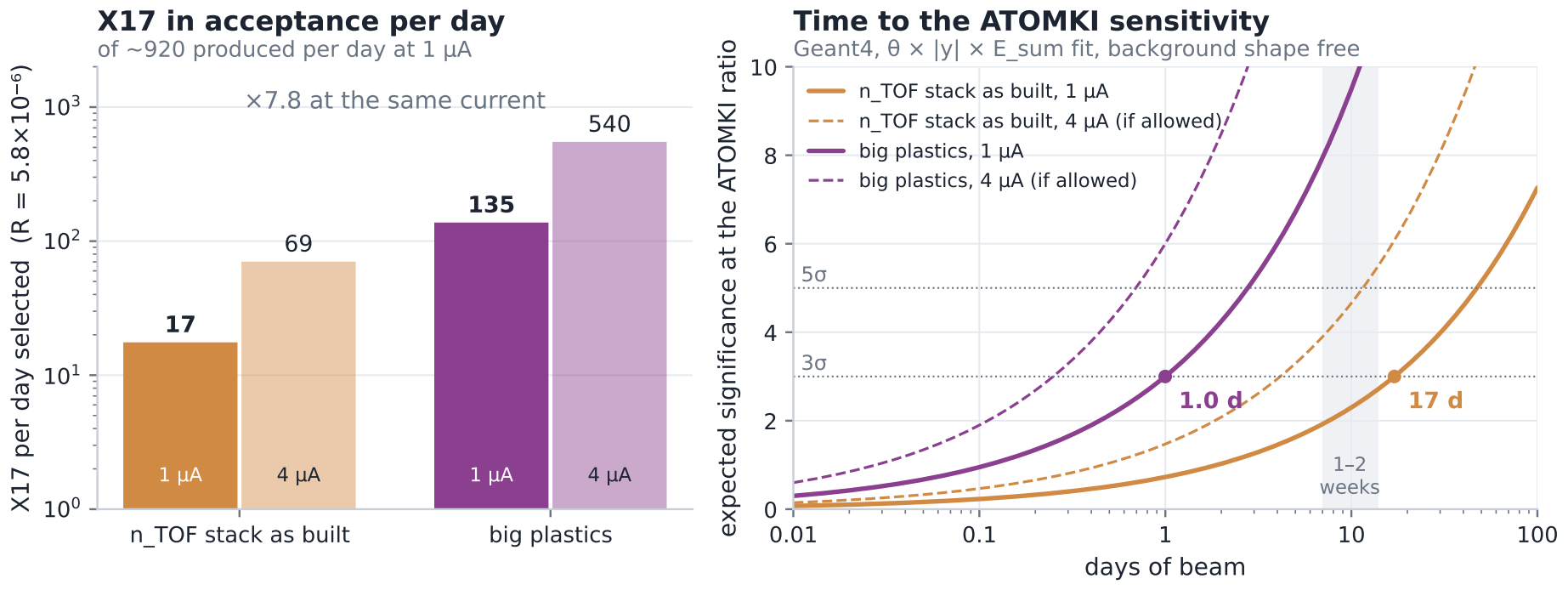
X17 in acceptance per day
of ~920 produced per day at 1 μA
×7.8 at the same current
X17 per day selected (R = 5.8×10⁻⁶)
10³
10²
10¹
10⁰
17
69
135
540
1 μA
4 μA
1 μA
4 μA
n_TOF stack as built
big plastics
Time to the ATOMKI sensitivity
Geant4, θ × |y| × E_sum fit, background shape free
n_TOF stack as built, 1 μA
n_TOF stack as built, 4 μA (if allowed)
big plastics, 1 μA
big plastics, 4 μA (if allowed)
5σ
3σ
1.0 d
17 d
1–2
weeks
expected significance at the ATOMKI ratio
0
2
4
6
8
10
0.01
0.1
1
10
100
days of beam